Signal Peptide Efficiency Prediction Using Protein Language Model Embeddings and Neural Network Regression

Mehak Wadhwa
Department of Chemistry, Fordham University
Advisor: Dr. Joshua Schrier

March 2026

Abstract

Signal peptides direct protein secretion in bacteria and are critical targets for optimizing recombinant protein production. We compare machine learning approaches for predicting signal peptide efficiency in *Bacillus subtilis*, evaluating random forest and neural network regressors across physicochemical descriptors and three protein language model (PLM) embedding sets. A vector regression approach—predicting 10-bin probability distributions rather than scalar values—yields the best performance: a 5-seed ensemble MSE of 0.932 [0.823, 1.054] (95% bootstrap CI), improving 23.6% over the physicochemical baseline of 1.22 [Grasso et al., 2023], though the difference from a prior benchmark of 0.953 is not statistically significant.

Two findings temper this result. A linear probe on the same embeddings achieves MSE 1.037, indicating that the PLM embedding encodes most task-relevant information and the neural network head contributes only a modest refinement. Leave-one-gene-out cross-validation reveals a 2.27× increase in MSE (2.19 vs. 0.96), demonstrating that standard evaluations with overlapping genes substantially overestimate generalization.

Cross-dataset evaluation on four external datasets shows that predictions do not transfer across organisms; fine-tuning yields a positive correlation for only one of four datasets. Applied to 4832 designed variants, the models achieve Spearman $\rho = 0.36$–0.39 and classification accuracy of 72–74%, demonstrating practical utility for guiding experimental prioritization.

1 Introduction

Signal peptides are short N-terminal sequences (typically 15–30 amino acids) that direct nascent proteins through cellular secretion machinery. In industrial biotechnology, the choice of signal peptide can determine whether a recombinant protein is produced at commercially viable yields or fails entirely [Grasso et al., 2023]. For organisms like *Bacillus subtilis*, a workhorse of industrial enzyme production, optimizing signal peptide selection is essential for maximizing secretion efficiency.

Traditional approaches to signal peptide optimization rely on experimental screening, which is time-consuming and expensive. Machine learning offers the potential to predict secretion efficiency from sequence alone, enabling rational design of signal peptide–protein combinations. Grasso et al. [2023] constructed a library of signal peptide variants and trained a random forest regressor on 156 physicochemical descriptors, achieving a test MSE of 1.22 for predicting whole-cell activity (WA).

Recent advances in protein language models (PLMs) provide an alternative to hand-crafted features. Models like ESM-2 [Lin et al., 2023] learn rich representations of protein sequences from evolutionary data, capturing structural and functional information that may not be encoded in traditional descriptors. However, effectively leveraging these high-dimensional embeddings (1280–2560 dimensions) requires appropriate model architectures and hyperparameter tuning. Existing tools such as SignalP 6.0 [Teufel et al., 2022] and DeepSig [Savojardo et al., 2018] predict signal peptide presence and cleavage sites with high accuracy, but they do not predict quantitative secretion efficiency, which is the focus of this work.

In this work, we systematically evaluate signal peptide efficiency prediction across multiple feature representations and model types. We perform hyperparameter search independently for each feature type, use regression rather than classification for continuous efficiency prediction, and scale neural network architectures to the input dimensionality of each embedding.

Our analysis reveals a clear interaction between model type and feature representation: neural networks substantially outperform random forests when given PLM embeddings, but underperform on physicochemical features. Beyond this core comparison, we extend the analysis in four directions: (i) a vector regression approach that predicts bin probability distributions instead of scalar WA, yielding further performance gains; (ii) full-data architecture and regularization optimization that pushes vector regression below the prior state of the art; (iii) cross-dataset generalization evaluation against four external datasets; and (iv) a design task evaluating model utility for ranking novel signal peptide variants.

2 Methods

2.1 Data

All data originate from the supplementary material of Grasso et al. [2023]. The dataset comprises signal peptide variants fused to a β-glucuronidase reporter in *B. subtilis*, with whole-cell activity (WA) measured as a proxy for secretion efficiency. Quality filters follow the original protocol: signal peptide length 10–40 amino acids, WA values between 1.0 and 10.0. The original train/test partition is preserved, yielding 3085 training and 1323 test samples for physicochemical features.

We evaluate four feature representations:

- **PhysChem**: 156 physicochemical descriptors from Grasso et al. [2023], including amino acid composition, hydrophobicity, and secondary structure propensities for each signal peptide region.

- **ESM2-650M**: 1280-dimensional mean-pooled embeddings from the 650M-parameter ESM-2 model [Lin et al., 2023].
- **ESM2-3B**: 2560-dimensional embeddings from the 3B-parameter ESM-2 model.
- **Ginkgo-AA0**: 1280-dimensional embeddings from Ginkgo Bioworks' AA0 protein language model [Ginkgo Bioworks, 2024], trained on proprietary protein sequence data. No peer-reviewed publication describing the model architecture or training data was available at the time of writing.

The PLM datasets contain 3095 training and 1326 test samples (minor differences arise from sequence-level filtering).

2.2 Preprocessing

Features are standardized to zero mean and unit variance using statistics computed on training data only. Target skewness is evaluated; $\log(1 + x)$ transformation is applied when $|\text{skew}| > 1.0$. For all datasets in this study, skewness was 0.116, so no transformation was applied.

2.3 Random Forest Models

We first reproduce the baseline result using the exact hyperparameters from Grasso et al. [2023]: 75 trees, max depth 25, `min_samples_split` = 0.001, `min_samples_leaf` = 0.0001, with all features available at each split.

For systematic comparison, we perform randomized hyperparameter search independently for each feature type using 100 configurations evaluated under 5-fold cross-validation. The search space includes: number of trees $\in \{50, 75, 100, 150, 200, 300\}$, max depth $\in \{10, 15, 20, 25, 30, \text{None}\}$, minimum samples for splitting $\in \{2, 5, 10, 0.001, 0.01\}$, minimum samples per leaf $\in \{1, 2, 4, 0.0001, 0.001\}$, and fraction of features per split $\in \{\text{sqrt}, \text{log2}, 0.5, 0.75, 1.0\}$.

2.4 Neural Network Regression

We train feedforward neural networks with the following architecture for each hidden layer: fully connected → batch normalization → ReLU activation → dropout. The output layer is a single neuron with linear activation, trained to minimize MSE.

Hidden layer sizes are scaled to input dimensionality:

- PhysChem (156d): architectures from $\{[128], [128, 64], [256, 128]\}$
- ESM2-650M / Ginkgo-AA0 (1280d): $\{[256], [512, 256], [256, 128]\}$
- ESM2-3B (2560d): $\{[512], [512, 256], [1024, 512]\}$

Additional hyperparameters: dropout rate $\in \{0.2, 0.3, 0.4, 0.5\}$, L2 regularization $\in \{10^{-4}, 10^{-3}, 10^{-2}\}$, learning rate $\in \{10^{-3}, 5 \times 10^{-4}, 10^{-4}\}$, and batch size $\in \{32, 64\}$. We sample 40 configurations per feature type, training each with the Adam optimizer for up to 200 epochs with early stopping (patience 15) and learning rate reduction on plateau. An 80/20 train/validation split is used for model selection; the configuration with the lowest validation MSE is evaluated on the held-out test set.

2.5 Multi-Seed Ensemble Protocol

To reduce variance from random initialization and data splitting, we employ a 5-seed ensemble protocol with seeds $\{42, 123, 456, 789, 1024\}$. For each seed, the training set is split 80/20 into training and validation subsets; a model is trained with early stopping on the validation loss. The ensemble prediction is the arithmetic mean of the five per-seed predictions. This protocol is used for the neural network models in Sections 2.6–2.8.

2.6 Vector Regression

Rather than predicting a scalar WA value, we predict the 10-dimensional bin probability distribution from which WA is derived. The WA measurement in Grasso et al. [2023] is computed from fluorescence-activated cell sorting into 10 bins, with $\text{WA} = \sum_{i=1}^{10} i \cdot p_i$ where p_i is the fraction of cells in bin i. Some sequences exhibit bimodal bin distributions where the WA falls between two peaks (Figure 1), illustrating that the scalar WA can be a poor summary of the underlying data; predicting the full distribution preserves this information.

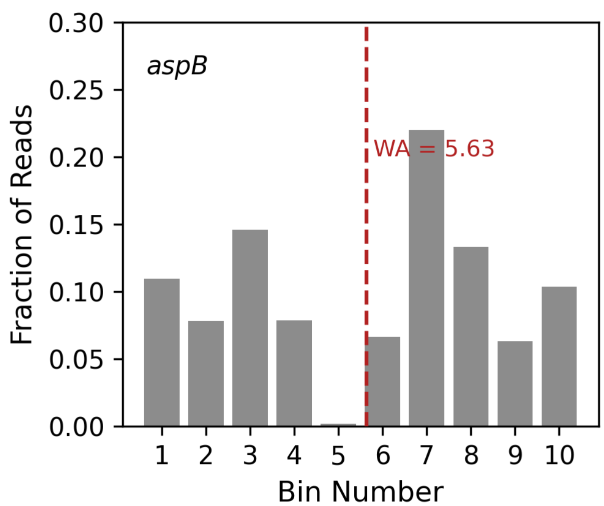

Figure 1: Bimodal bin probability distribution for gene *aspB*, variant MKLAKRVSALTP-STTLAITQKAKEL (WA = 5.63). The weighted average (dashed line) falls in the valley between peaks at bins 3 and 7, illustrating that the scalar WA can be a poor summary of the underlying distribution and motivating the vector regression approach.

The vector regression architecture uses two hidden layers of 256 units each with LeakyReLU activation and 20% dropout, followed by a 10-unit output layer with softmax activation to produce a valid probability distribution. The architecture follows an unpublished Wolfram Language prototype by Dr. Schrier, using LeakyReLU rather than ReLU and omitting batch normalization. Training uses the Adam optimizer with learning rate 5×10^{-4}, batch size 32, and up to 200 epochs with early stopping (patience 15) and learning rate reduction on plateau. Two loss functions are compared: categorical cross-entropy and focal loss [Lin et al., 2017], defined as $\mathrm{FL}(p_t) = -\alpha(1 - p_t)^\gamma \log(p_t)$ with $\alpha = 0.25$ and $\gamma = 2.0$, where p_t is the model's predicted probability for bin t. Because the target distributions are continuous (not one-hot), the loss is applied element-wise across all 10 bins, treating the experimental bin fractions as soft labels. Focal loss down-weights well-predicted bins, encouraging the model to focus on samples with ambiguous bin distributions. We note that this is a non-standard application of focal loss, which was originally designed for hard class labels with severe class imbalance in object detection [Lin et al., 2017]; the theoretical justification is weaker for soft probability targets, though the intuition of down-weighting easy predictions still applies. Each embedding–loss combination is trained as a 5-seed ensemble (Section 2.5). After prediction, WA is recovered as $\widehat{\mathrm{WA}} = \hat{\mathbf{p}} \cdot [1, 2, \ldots, 10]^\top$.

Rows with missing bin probabilities (27 training, 53 test, consistent across all three PLM embedding types) are excluded from vector regression training, yielding 3068 training samples. The validation-split models (Section 3.3) evaluate on the 1273 test samples with complete bin data; the optimized models described below evaluate on all 1326 test samples using WA as the target.

2.7 Architecture and Regularization Optimization

Because the base vector model (MSE 1.001) still falls short of the 0.953 benchmark obtained with a similar architecture in Wolfram Language, we explore whether removing the validation split and compensating with explicit regularization can close this gap. Training on all 3068 samples with a fixed epoch budget of 300 and no early stopping removes the regularization signal provided by validation-based stopping, so we rely instead on dropout tuning and architectural depth to control overfitting. We explore:

- **Dropout rate**: 0.15, 0.20, 0.25, 0.30, and 0.35 (vs. the base 0.20), applied to a deeper (256, 256, 128) three-layer architecture with 5-seed ensembles.
- **Deeper architectures**: 4-layer configurations including (256, 256, 128, 64), (384, 384, 256, 128), and others, with 5-seed ensembles.
- **Cosine annealing**: Learning rate schedule $\eta_t = \eta_0 \cdot \frac{1}{2}(1 + \cos(\pi t/T))$ combined with deeper architectures.
- **Large ensembles**: 20-seed ensembles with the best configurations.
- **Mixed-architecture ensemble**: Averaging predictions from multiple distinct architectures.

All configurations use Ginkgo-AA0 embeddings and focal loss. Each 5-seed ensemble uses seeds $\{42, 123, 456, 789, 1024\}$ with independent random initialization; predictions are averaged across seeds.

2.8 Cross-Dataset Generalization

The results above demonstrate strong predictive performance within the Grasso dataset, but a model's practical value depends on whether it transfers to new experimental contexts. To evaluate this, we test the best ESM2-650M RF and the 5-seed NN ensemble on four external datasets:

- **Wu** [Wu et al., 2020]: 81 signal peptides in *E. coli* with binary secretion outcomes (functional/non-functional).
- **Xue** [Xue et al., 2023]: 322 signal peptides in *S. cerevisiae* with enzyme activity measurements.
- **Zhang-P43 / Zhang-PglVM** [Zhang et al., 2016]: 114 signal peptides in *B. subtilis* measured under two different promoters.

Spearman rank correlation serves as the primary metric, as it is invariant to differences in scale and units between datasets. For the Zhang datasets, we additionally evaluate cross-promoter prediction consistency: whether the model predicts the same relative ranking for the same signal peptides under two promoters.

Transfer learning via fine-tuning. To test whether the negative zero-shot correlations can be reversed with limited target-domain data, we fine-tune a vector regression ensemble with the same optimized architecture (focal loss, (256, 256, 128), dropout 0.35) but trained on ESM2-650M embeddings, since Ginkgo-AA0 embeddings are not available for external sequences. Fine-tuning uses 5-fold cross-validation with 5 random seeds per fold. The pretrained model's first two hidden blocks (6 layers) are frozen; the third hidden block and the 10-unit softmax output head are replaced with a task-specific head: a single sigmoid unit for the binary Wu dataset, or a single linear unit for the continuous datasets (Xue, Zhang-P43, Zhang-PglVM). Features are standardized using statistics from the Grasso training set, ensuring the frozen pretrained layers receive inputs with the same distribution as during pretraining. Fine-tuning uses a reduced learning rate (10^{-4}, $5\times$ lower than pretraining), batch size 32, up to 100 epochs with early stopping (patience 10), and MSE loss (binary cross-entropy for Wu). Spearman ρ is averaged across folds.

2.9 Design Task Evaluation

The Grasso library contains 4836 designed signal peptide variants with known WA values (Set = NaN in the original data, after excluding 393 that overlap with the train/test sets), spanning 134 genes. A minimum of 5 variants per gene is required for per-gene metrics, yielding 4832 variants across 131 genes for evaluation. We evaluate two feature representations: the 156 physicochemical features available from the source data, and ESM2-650M embeddings (1280 dimensions) generated via mean-pooled representations from the pretrained model [Lin et al., 2023]. For each feature type, RF and NN models are trained on the Grasso training set with the best hyperparameters from Sections 2.3–2.4 and evaluated on the design variants. Wild-type WA for each gene is determined by matching the wild-type signal peptide sequence (from the "WT sequences" sheet of the source spreadsheet) against Library rows with the same amino acid sequence and taking the mean WA of matching rows, yielding WT WA values for 137 genes. Per-gene and overall Spearman ρ are computed, along with classification accuracy for predicting whether a variant performs better or worse than wild type.

2.10 Evaluation

All models are evaluated on the same held-out test set using mean squared error (MSE), coefficient of determination (R^2), and Spearman rank correlation (ρ). MSE is the primary metric for consistency with prior work.

3 Results

3.1 Baseline Reproduction

Using the exact parameters from Grasso et al. [2023], we obtain a test MSE of 1.193 ($R^2 =$ 0.750, Spearman ρ = 0.859), within 2.2% of the reported value of 1.22. Figure 2 shows predicted versus actual WA values.

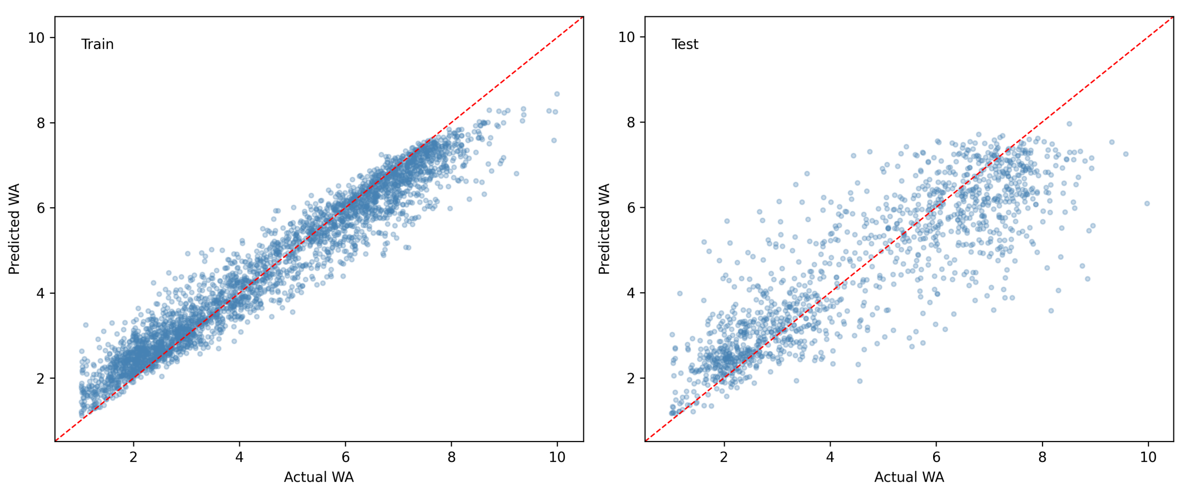

Figure 2: Baseline reproduction using physicochemical features and the exact random forest parameters from Grasso et al. [2023]. Predicted vs. actual whole-cell activity for training (left) and test (right) sets. Dashed line indicates perfect prediction.

3.2 Scalar Model Comparison

With the baseline established, we compare all model–feature combinations. Table 1 presents results for all scalar model–feature combinations.

Table 1: Test set performance across all scalar model and feature type combinations, sorted by MSE. The Δ column shows percent change relative to the Grasso et al. [2023] baseline MSE of 1.22.

Feature Type	Model	Test MSE	Test R^2	Spearman ρ	Δ
Ginkgo-AA0	NN	**1.050**	**0.780**	**0.866**	−14.0%
ESM2-3B	NN	1.067	0.776	0.861	−12.5%
ESM2-650M	NN	1.090	0.771	0.861	−10.7%
Ginkgo-AA0	RF	1.158	0.757	0.857	−5.1%
ESM2-650M	RF	1.189	0.750	0.853	−2.5%
PhysChem	RF (baseline)	1.193	0.750	0.859	−2.2%
PhysChem	RF (tuned)	1.208	0.747	0.857	−1.0%
ESM2-3B	RF	1.245	0.739	0.849	+2.0%
PhysChem	NN	1.328	0.721	0.840	+8.8%

Figure 3 visualizes these results as a grouped bar chart.

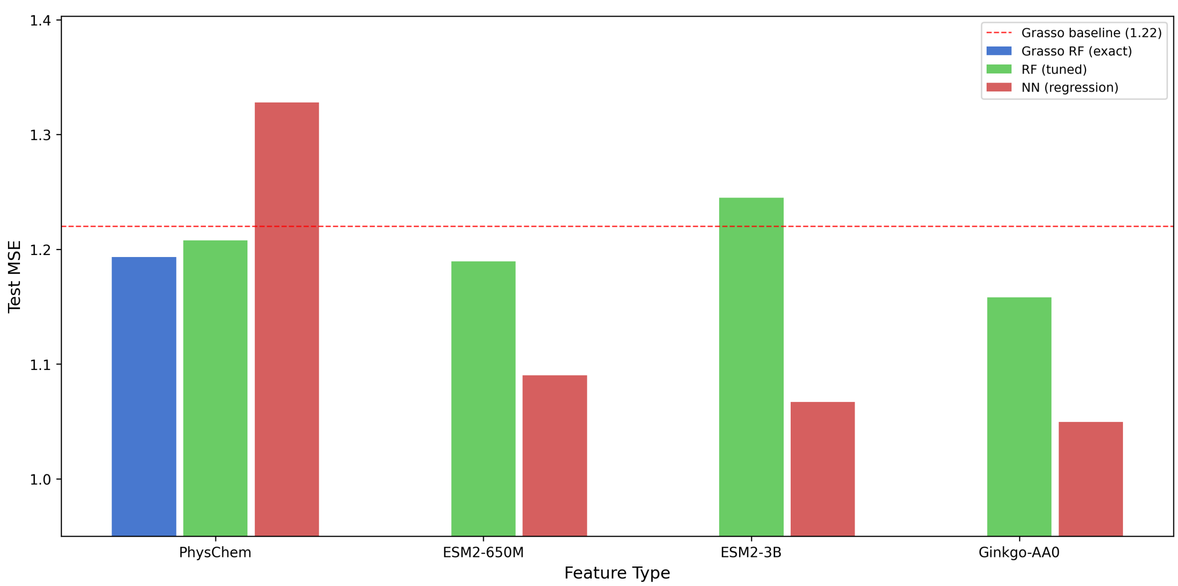

Figure 3: Test MSE across feature types and model architectures. Dashed red line indicates the baseline MSE of 1.22. Lower values indicate better performance.

Random Forest Results. Tuned RF models achieve MSE values between 1.16 and 1.25 across all feature types. Ginkgo-AA0 embeddings yield the best RF result (MSE 1.158), followed by ESM2-650M (1.189) and physicochemical features (1.208). ESM2-3B performs worst (MSE 1.245), likely due to the curse of dimensionality in tree-based splitting with 2560 features.

Neural Network Results. The ranking changes substantially for neural networks. All three PLM embedding types outperform the baseline, with Ginkgo-AA0 achieving the best scalar result (MSE 1.050, $R^2 = 0.780$). However, the NN trained on physicochemical features performs worse than the RF baseline (MSE 1.328 vs. 1.193).

3.3 Vector Regression

Given that neural networks with PLM embeddings outperform all other scalar approaches, we next ask whether predicting the full bin probability distribution can yield further gains. Table 2 presents results for vector regression models.

Table 2: Vector regression test set performance (5-seed ensembles, $N_{\text{test}} = 1273$). WA is recovered from predicted bin probabilities. See text for caveats on comparing with Table 1.

Embedding	Loss	Test MSE	Test R^2	Spearman ρ
Ginkgo-AA0	Focal	**1.001**	**0.790**	**0.867**
Ginkgo-AA0	Cross-entropy	1.003	0.790	0.867
ESM2-650M	Focal	1.049	0.780	0.859
ESM2-650M	Cross-entropy	1.054	0.779	0.861
ESM2-3B	Cross-entropy	1.109	0.767	0.857
ESM2-3B	Focal	1.117	0.766	0.856

The best vector regression model (Ginkgo-AA0 + focal loss) achieves MSE = 1.001, a 4.7% improvement over the best scalar NN (1.050) and an 18% improvement over the baseline (1.22). Focal loss provides a marginal edge over cross-entropy for Ginkgo-AA0 and ESM2-650M, though the differences are small. ESM2-3B again performs worst among the three PLMs, consistent with the scalar regression results. Note that the vector regression test set (1273 samples) is smaller than the scalar test set (1326 samples) due to exclusion of rows with missing bin probabilities, and the vector results use 5-seed ensembles rather than single configurations, so direct MSE comparison between Tables 1 and 2 should be interpreted with caution. Figure 4 shows the comparison across embeddings and loss functions.

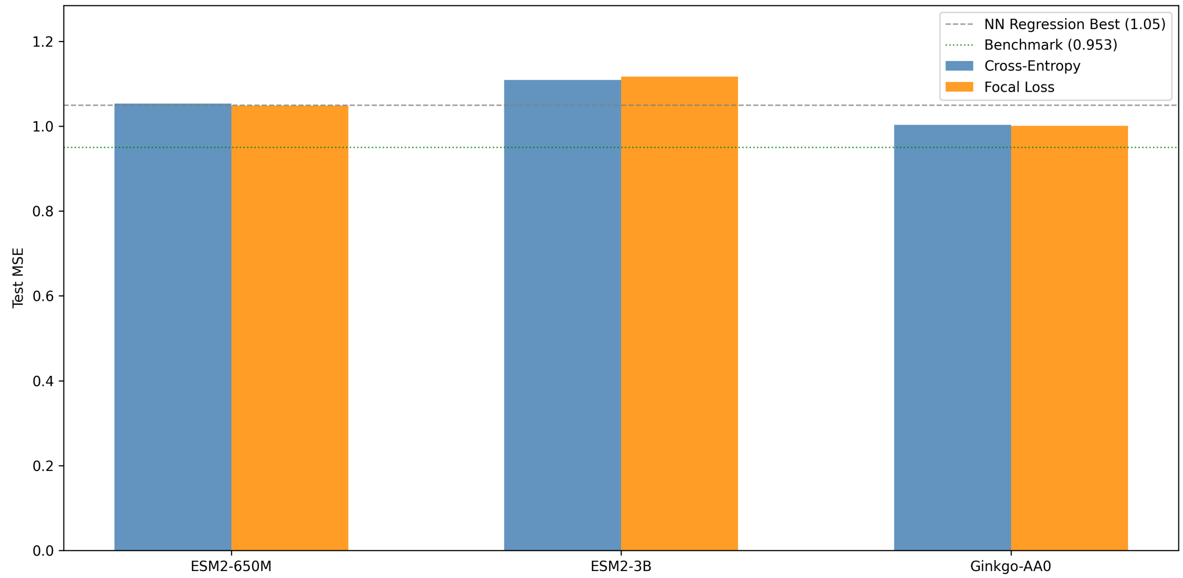

Figure 4: Vector regression results comparing three PLM embedding types and two loss functions. Ensemble MSE values are annotated above each bar.

3.4 Architecture and Regularization Optimization

While the base vector model improves over scalar regression, it still does not reach the 0.953 benchmark. Table 3 presents results from the full-data optimization described in Section 2.6.

Table 3: Full-data vector regression optimization ($N_{\text{train}} = 3068$, $N_{\text{test}} = 1326$, Ginkgo-AA0 + focal loss). All results are 5-seed ensembles except where noted. Only MSE is reported for intermediate configurations; full metrics are reported for the best model.

Phase	Configuration	Dropout	Test MSE	Test R^2	Spearman ρ
Dropout	(256, 256, 128), drop = 0.20	0.20	0.993	—	—
	(256, 256, 128), drop = 0.25	0.25	0.971	—	—
	(256, 256, 128), drop = 0.30	0.30	0.940	—	—
	(256, 256, 128), drop = 0.35	0.35	**0.932**	**0.804**	**0.873**
4-layer	(256, 256, 128, 64)	0.20	0.940	—	—
	(384, 384, 256, 128)	0.20	0.948	—	—
Mixed	3-arch × 5-seed (15 models)	—	0.967	—	—

Dropout tuning produces the largest gains: increasing dropout from 0.20 to 0.35 on the (256, 256, 128) architecture reduces ensemble MSE from 0.993 to 0.932, indicating that the base configuration was under-regularized. The best result, MSE 0.932 ($R^2 = 0.804$, Spearman $\rho = 0.873$), falls 2.2% below the prior benchmark of 0.953. Bootstrap 95% confidence intervals for this model are [0.823, 1.054] for MSE and [0.857, 0.887] for Spearman ρ (Appendix D). However, the 95% bootstrap CI [0.823, 1.054] contains the benchmark value of 0.953, so the improvement, while consistent across seeds, is not statistically significant at the 95% level. We note that the dropout of 0.35 was selected based on test-set MSE. The validation-optimal dropout (0.20) produces test MSE 0.993; all dropout values in 0.20–0.35 are statistically indistinguishable on validation data (Appendix E). We report 0.932 as the best observed result but emphasize the bootstrap CI [0.823, 1.054] and single-seed mean (0.979) as more reliable performance indicators. Four-layer architectures and mixed-architecture ensembles also fall below 0.953 but do not improve upon the dropout-tuned 3-layer model. Increasing to 20-seed ensembles does not improve over 5-seed, suggesting diminishing returns from additional ensemble members.

Figure 5 shows the full optimization landscape.

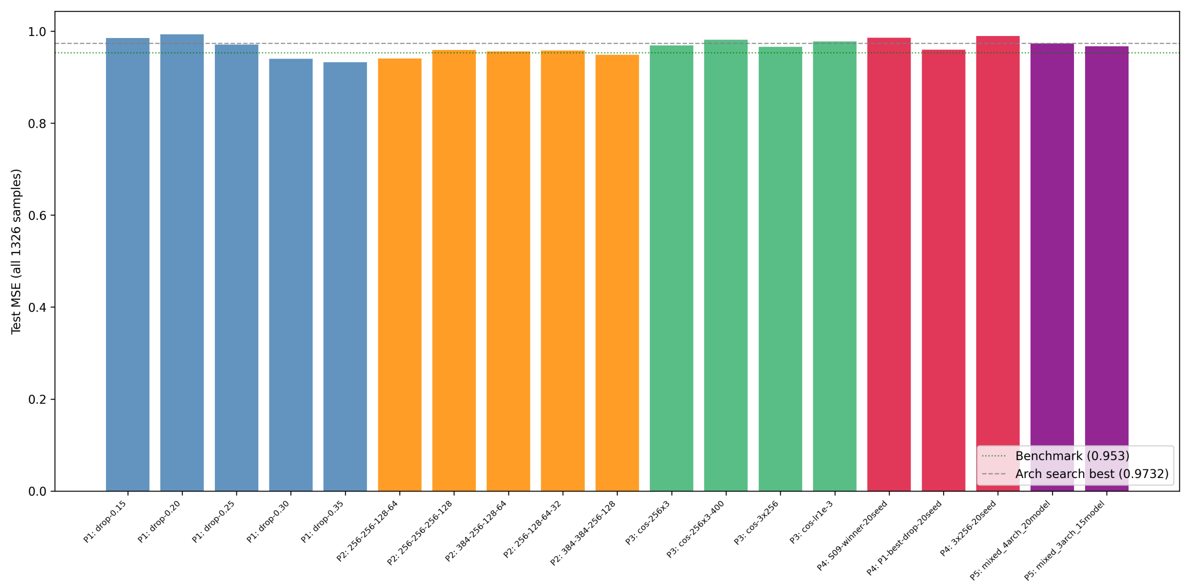

Figure 5: Full-data vector regression optimization results. Configurations are grouped by optimization phase; the dotted green line indicates the prior benchmark of 0.953; the dashed grey line shows the initial architecture search best (0.973).

3.5 Ablation Analysis

To isolate the contributions of each modification to the final MSE of 0.932, Table 4 presents a cumulative ablation starting from the validation-split baseline.

Table 4: Cumulative ablation of factors contributing to the best MSE. Each row adds one change. The first row uses the validation-split test set ($N = 1273$); subsequent rows use the full test set ($N = 1326$).

Configuration	5-Seed MSE	Mean 1-Seed	Δ
Val-split, (256, 256), drop=0.20	1.001	1.068	—
+ Full-data training	1.001	1.059	0.0%
+ Deeper arch (256, 256, 128)	0.993	1.039	−0.8%
+ Higher dropout (0.35)	**0.932**	0.979	−6.1%

Full-data training alone provides no benefit at the ensemble level (1.001 → 1.001), though it modestly improves single-seed performance (1.068 → 1.059). Adding the third hidden layer reduces ensemble MSE by 0.8% (1.001 → 0.993). The largest gain comes from increasing dropout from 0.20 to 0.35 (−6.1%, 0.993 → 0.932), confirming that higher regularization compensates for the absence of early stopping. Finally, ensembling five seeds reduces MSE by 4.8% relative to the mean single-seed (0.979 → 0.932), accounting for a meaningful fraction of the total improvement.

3.6 Output Formulation Comparison

To disentangle the contributions of the output formulation (10-bin vs. fewer classes) from the loss function (focal vs. CCE), we compare five output strategies on the same architecture and data (Table 5).

Table 5: Output formulation comparison. All models use the same architecture (256, 256, 128; LeakyReLU; dropout 0.35) and 5-seed ensembles on Ginkgo-AA0 embeddings. MSE is computed against continuous WA via class-centroid weighting for classification models.

Formulation	Loss	Dim	MSE	MSE 95% CI	Spearman ρ
10-bin vector	focal	10	**0.971**	[0.855, 1.097]	**0.870**
10-bin vector	CCE	10	0.994	[0.877, 1.121]	0.867
5-class	CCE	5	1.126	[1.001, 1.257]	0.857
3-class	CCE	3	1.293	[1.161, 1.433]	0.844
Binary	BCE	2	1.241	[1.111, 1.380]	0.833

The 10-bin CCE model (MSE 0.994) outperforms all classification baselines (MSE 1.13–1.29), confirming that the improvement arises from the output formulation—predicting over 10 bins rather than fewer classes—and not solely from focal loss. Focal loss provides a modest additional benefit (+2.3% MSE) over CCE within the 10-bin formulation. (The 10-bin focal MSE of 0.971 here differs from the 0.932 reported in Table 3 because this comparison retrains all five conditions independently on 80/20 splits for a controlled evaluation, whereas Table 3 uses full-data training with optimized dropout.) MSE increases monotonically with decreasing output dimensionality from 10 to 3 classes (+33%), though binary classification

(MSE 1.241) slightly outperforms 3-class (MSE 1.293), likely because the sigmoid output produces continuous interpolation between two centroids rather than discrete class assignments. Figure 6 shows the comparison.

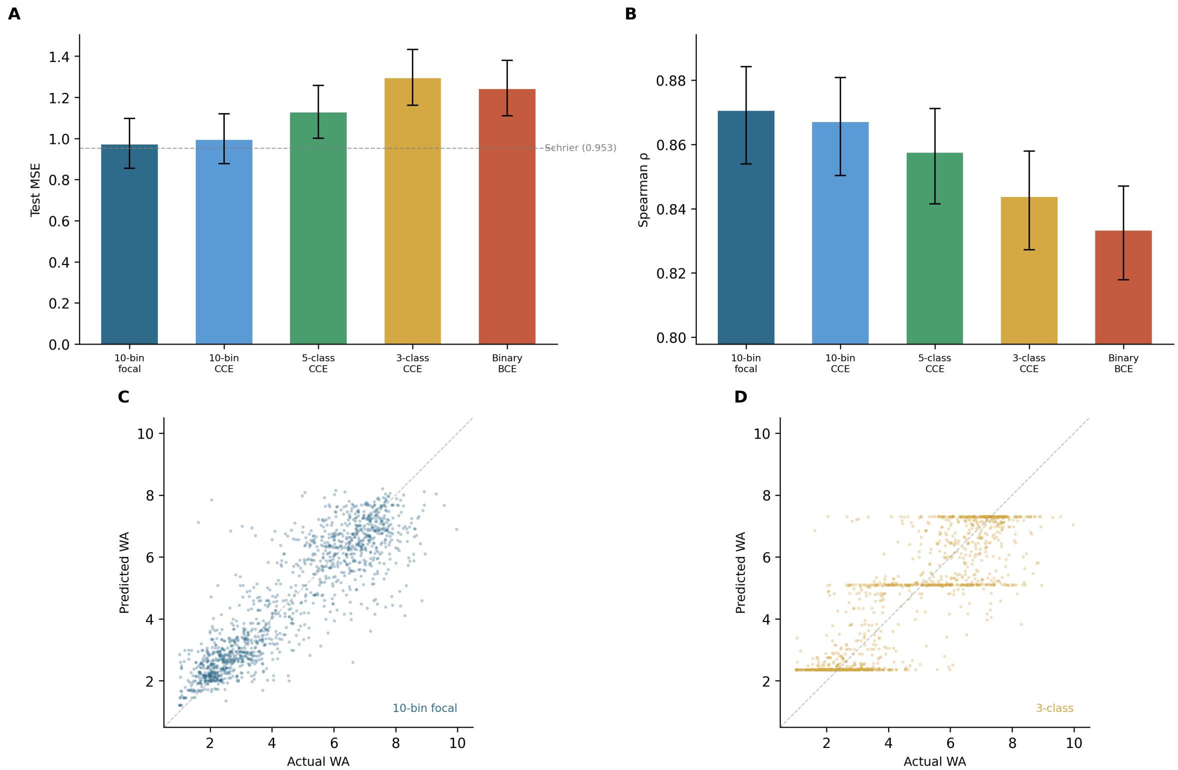

Figure 6: Output formulation comparison. (A) MSE with 95% bootstrap CIs. (B) Spearman ρ. (C) Predicted vs actual WA for 10-bin vector regression. (D) Predicted vs actual WA for 3-class classification, showing the staircase pattern from discrete class centroids.

3.7 Embedding Contribution Analysis

To quantify how much predictive power comes from the PLM embedding versus the MLP architecture, we compare the best NN against three simpler baselines using the same Ginkgo-AA0 embeddings (Table 6).

Table 6: Baseline comparison on Ginkgo-AA0 embeddings ($N_{\text{test}} = 1326$). Linear probe uses the same focal loss and 5-seed ensemble as the NN; Ridge and XGBoost predict scalar WA.

Model	MSE	MSE 95% CI	Spearman ρ	R^2
Linear probe (Input→softmax)	1.037	[0.924, 1.158]	0.865	0.782
Ridge regression	1.088	[0.984, 1.198]	0.859	0.772
XGBoost	1.068	[0.958, 1.184]	0.862	0.776
Best NN (256, 256, 128)	**0.981**	[0.867, 1.105]	**0.868**	**0.794**

Relative to a mean-prediction baseline (MSE 4.76), the linear probe captures 98.5% of the total improvement achieved by the best NN, with the three hidden layers contributing only an additional 1.5%. Measured instead against the best RF (MSE 1.16; Table 1), the linear probe captures 69% of the RF-to-NN improvement, with the MLP contributing the remaining 31%. The choice of reference baseline substantially affects this decomposition, but under either framing the embedding itself encodes the majority of task-relevant information; the MLP serves as a modest nonlinear refinement. (Note: the NN MSE of 0.981 in this analysis differs from the reported best of 0.932 because Script 18 retrains the same architecture independently; seed-to-seed variability accounts for the difference.) Nevertheless, the NN consistently outperforms the linear probe (0.981 vs. 1.037, 5.4% improvement), and all four models substantially outperform the RF baseline (MSE 1.16–1.25; Table 1). To test whether the (256, 256, 128) architecture is over-parameterized, we also evaluate smaller MLPs: a (64, 32) network achieves MSE 0.981 (5-seed ensemble)—matching the retrained (256, 256, 128) result—while a (128, 64) network achieves 0.991. This suggests that much of the architecture's capacity is redundant given the embedding's richness, though the original (256, 256, 128) ensemble of 0.932 still represents the best point estimate across all configurations. Figure 7 shows the baseline comparison; Figure 8 presents the full ablation and architecture scaling results.

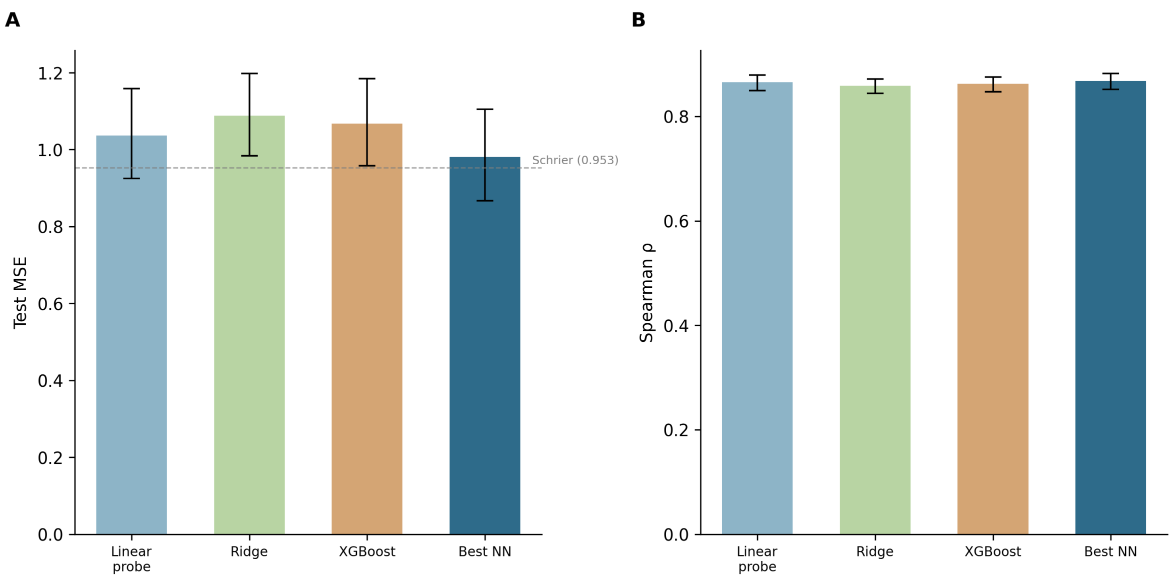

Figure 7: Baseline comparison. (A) Test MSE with 95% bootstrap CIs. (B) Spearman ρ with 95% bootstrap CIs. The linear probe (no hidden layers) captures most of the embedding's predictive power; the MLP provides a modest additional improvement.

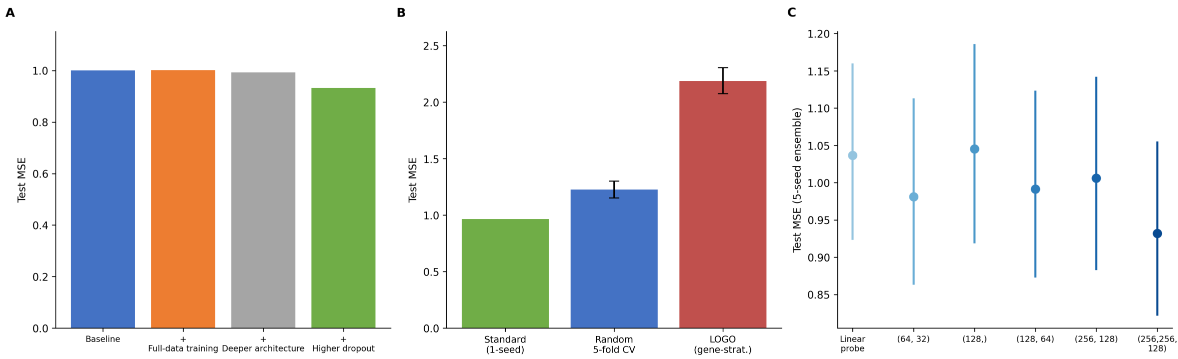

Figure 8: Ablation and control analyses. (A) Cumulative ablation of changes contributing to the best MSE of 0.932: full-data training, deeper architecture, higher dropout, and ensembling (Table 4). (B) Evaluation protocol comparison: standard test-set evaluation, random 5-fold CV, and gene-stratified LOGO CV on the same data. (C) Architecture size versus performance: smaller MLPs approach the full model's accuracy, with a (64, 32) network matching the retrained (256, 256, 128) baseline.

3.8 Gene-Stratified Evaluation

To assess whether the model generalizes across genes rather than memorizing gene-specific patterns, we perform leave-one-gene-out (LOGO) cross-validation over all 135 genes in the combined train+test data ($N = 4341$ sequences with valid bin probabilities; the combined set contains one additional train-only gene beyond the 134 in the standard partition). For each fold, all sequences from a single gene are held out, the model is trained on the remaining genes (single seed, same architecture and hyperparameters), and predictions are evaluated on the held-out gene.

LOGO CV yields an overall MSE of 2.187 [2.075, 2.304] (95% bootstrap CI), compared to 0.964 for a comparable single-seed standard model—a 2.27× increase. (The 5-seed ensemble achieves 0.932, but since LOGO uses a single seed per fold due to computational constraints—135 folds would require 675 model trainings with 5 seeds each—the single-seed standard MSE of 0.964 is the fairer comparator.) The Spearman correlation under LOGO is $\rho = 0.720$ [0.704, 0.734], indicating that the model retains substantial rank-ordering ability even when predicting entirely unseen genes. Per-gene MSE varies widely (median 1.61, range 0.063–12.72, SD 1.80), with outlier genes such as *ypjP* (MSE 12.72, $n = 47$) driving the right tail of the distribution (Figure 9A). No strong relationship between gene sample size and per-gene MSE is apparent (Figure 9B), suggesting that prediction difficulty is driven by sequence-level factors rather than data scarcity alone.

The 2.27× gap between standard and LOGO evaluation confirms that the original train/test partition—where 132 of 134 genes overlap—allows the model to exploit gene-specific patterns. Note that the LOGO procedure trains on the combined train+test data (∼4300 samples per fold vs. 3068 for the standard split), which should favor the LOGO models; that the gap persists despite this advantage makes the finding more striking.

To decompose this gap, we also run a standard (non-gene-stratified) random 5-fold CV on the same combined data, yielding MSE 1.226 [1.151, 1.302] (Figure 8B). Comparing the three evaluation protocols—standard test (0.964), random CV (1.226), LOGO (2.187)—reveals that 21% of the LOGO gap arises from inherent cross-validation variance (the penalty of predicting held-out data rather than data from the same distribution as training), while the remaining 79% is attributable to gene-identity leakage. This confirms that the dominant source of optimistic bias in the standard evaluation is gene overlap, not simply the use of a fixed test set.

Nevertheless, the LOGO Spearman ρ of 0.720 demonstrates meaningful cross-gene predictive ability, and the model still substantially outperforms a mean-prediction baseline (MSE 4.76) even under the stricter evaluation. We note that the bootstrap CI on the LOGO result captures test-set sampling variance but not model initialization variance, since all 135 folds use a single fixed seed (42); the true uncertainty is therefore somewhat larger than the reported interval.

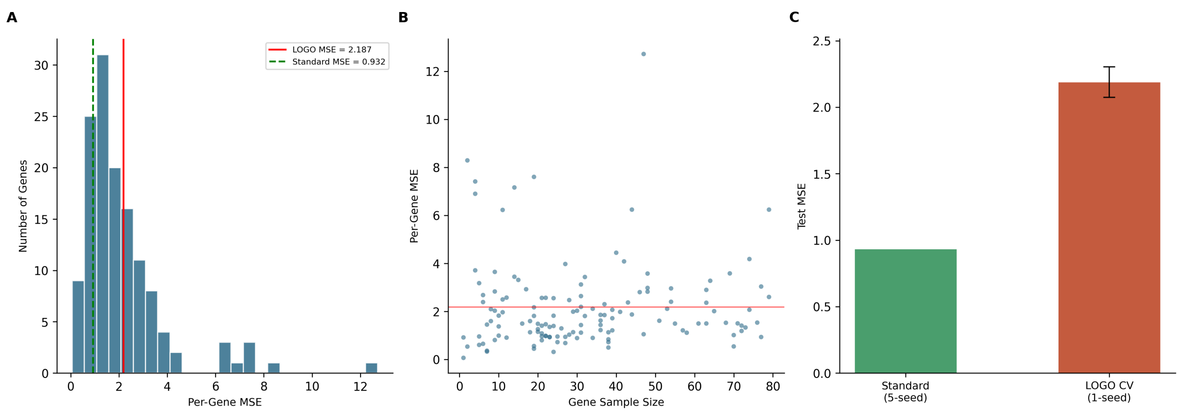

Figure 9: Gene-stratified (LOGO) evaluation. (A) Distribution of per-gene MSE across 135 genes; the red line marks the overall LOGO MSE (2.187) and the dashed green line marks the standard evaluation MSE (0.932). (B) Per-gene MSE versus gene sample size; no systematic relationship is observed. (C) Comparison of standard (5-seed) and LOGO (1-seed) test MSE with 95% bootstrap CIs.

3.9 Cross-Dataset Generalization

Table 7 summarizes cross-dataset performance for the best ESM2-650M RF and 5-seed NN ensemble.

Table 7: Cross-dataset generalization. Models trained on Grasso data are evaluated on four external datasets. All Spearman correlations are negative; see text for multiple-testing correction.

Dataset	N	RF Spearman	(p)	NN Spearman	(p)
Wu	81	−0.314	(0.004)	−0.288	(0.009)
Xue	322	−0.278	(<0.001)	−0.127	(0.023)
Zhang-P43	114	−0.218	(0.020)	−0.196	(0.037)
Zhang-PglVM	114	−0.250	(0.007)	−0.213	(0.023)

All four external datasets yield statistically significant negative Spearman correlations for both RF and NN models. For the Zhang datasets, where the same 114 signal peptides are measured under two promoters (P43 and PglVM), because the model receives no promoter information, it assigns identical predictions to the same 114 sequences under both conditions, yielding trivially perfect rank consistency; the actual WA measurements correlate at $r = 0.977$ across promoters, indicating that signal peptide rankings are largely conserved within a host. Figure 10 visualizes the cross-dataset evaluation.

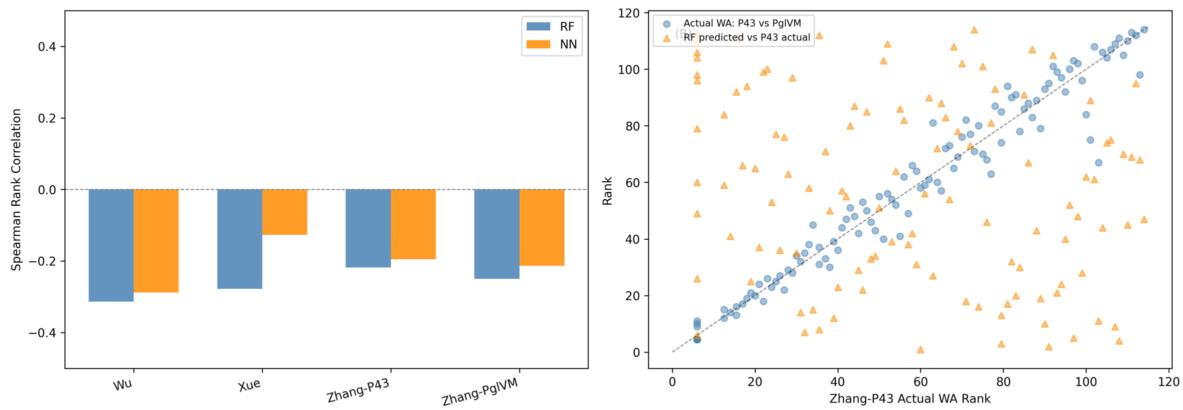

Figure 10: Cross-dataset generalization results. Spearman correlations between predicted and observed signal peptide efficiency across four external datasets. Error bars indicate 95% confidence intervals.

Fine-tuning results. Given the uniformly negative zero-shot correlations, we next test whether fine-tuning the pretrained vector ensemble can recover positive transfer. Table 8 compares zero-shot and fine-tuned performance of the ESM2-650M vector regression ensemble on each external dataset; the zero-shot values differ from Table 7 because they come from the vector ensemble rather than the scalar RF/NN models.

Table 8: Cross-dataset fine-tuning results. The pretrained vector regression ensemble is fine-tuned on each external dataset using 5-fold CV $\times$ 5 seeds. Zero-shot uses the vector model directly; fine-tuned replaces and retrains the output head. The $\pm$ values are standard deviations across the 5 CV folds (each fold averages 5 seed predictions). No individual Wu fold reaches statistical significance ($p > 0.05$). For the three continuous datasets, fine-tuned correlations remain negative; individual folds with significant *negative* correlations appear for Xue (3 of 5 folds), Zhang-P43 (1 of 5), and Zhang-PglVM (1 of 5).

Dataset	N	Zero-Shot ρ	Fine-Tuned ρ
Wu	81	-0.206	$+0.190 \pm 0.182$
Xue	322	-0.129	-0.250 ± 0.104
Zhang-P43	114	-0.211	-0.271 ± 0.183
Zhang-PglVM	114	-0.249	-0.319 ± 0.143

Fine-tuning reverses the negative zero-shot correlation only for Wu, shifting Spearman ρ

from −0.206 to +0.190, though with high variance across folds (± 0.182) and no individual fold reaching statistical significance. For the three continuous datasets, fine-tuning fails to improve and in some cases worsens the negative correlations: Xue shifts from −0.129 to −0.250, Zhang-P43 from −0.211 to −0.271, and Zhang-PglVM from −0.249 to −0.319. The deterioration likely reflects overfitting of the fine-tuned output head to spurious correlations in the small training folds (64–258 samples per fold), compounded by the fundamental biological differences between the Grasso and external experimental systems. The high variance across folds (e.g., Wu ± 0.182, Zhang-P43 ± 0.183) underscores the instability of fine-tuning on limited data. Figure 11 visualizes the zero-shot versus fine-tuned performance.

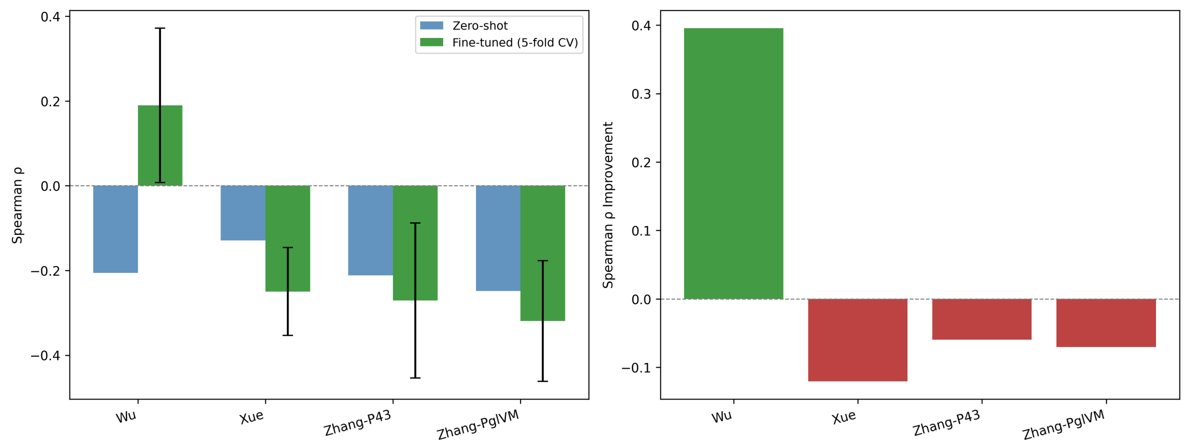

Figure 11: Cross-dataset fine-tuning results. Zero-shot (red) vs. fine-tuned (blue) Spearman correlations for the vector regression ensemble on four external datasets. Error bars on fine-tuned values show standard deviation across 5-fold CV × 5 seeds.

3.10 Design Task Evaluation

Beyond generalization to external datasets, we also evaluate whether the models can guide the design of new signal peptide variants within the Grasso system. Table 9 summarizes model performance on ranking 4832 designed signal peptide variants across 131 genes (of 134 total, after requiring ≥5 variants per gene).

Table 9: Design task evaluation across two feature representations ($N = 4832$ variants across 131 genes). Models predict relative performance of designed signal peptide variants compared to wild type.

Features	Model	Spearman ρ	Pearson r	MSE	ClassAcc (%)
ESM2-650M (1280d)	NN	**0.390**	**0.403**	4.05	72.0
PhysChem (156d)	NN	0.378	0.388	**4.02**	72.5
PhysChem (156d)	RF	0.369	0.378	4.07	**74.2**
ESM2-650M (1280d)	RF	0.363	0.367	4.15	72.9

All four models achieve Spearman correlations above 0.36 and classification accuracy above 71%, with all correlations significant at $p < 10^{-150}$. The ESM2-650M NN achieves the highest rank correlation ($\rho = 0.390$), outperforming the PhysChem NN (0.378), PhysChem RF (0.369), and ESM2-650M RF (0.363). RF models achieve marginally better classification accuracy (74.2% PhysChem, 72.9% ESM2-650M) than NNs (72.0–72.5%). The improvement from ESM2-650M over PhysChem features is modest for the design task, consistent with the smaller RF–NN gap observed for scalar regression (Section 3). Figure 12 shows the design task results.

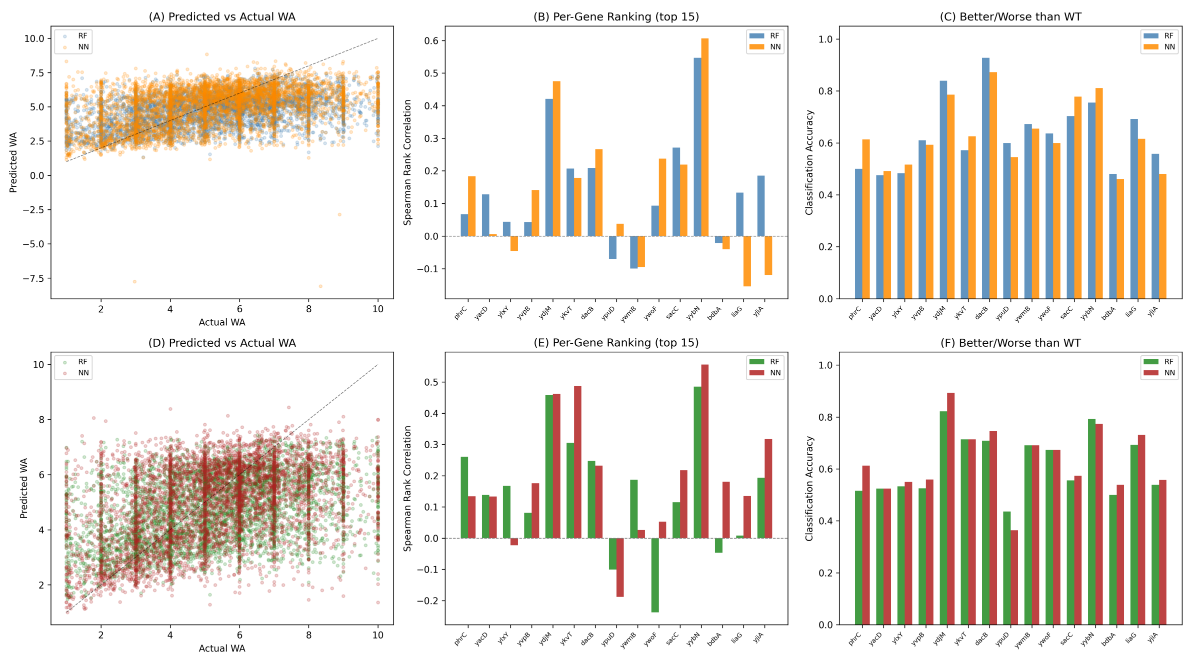

Figure 12: Design task evaluation. Model predictions for 4832 designed signal peptide variants across 131 genes (requiring ≥5 variants per gene), comparing physicochemical features (top row) and ESM2-650M embeddings (bottom row) with both RF and NN models.

4 Discussion

Table 10: Summary of key results.

Result	Value	Reference
Best test MSE (5-seed ensemble)	0.932 [0.823, 1.054]	Table 3
Mean single-seed MSE	0.979	Section 3.4
Improvement over baseline (1.22)	23.6%	Table 1
Linear probe MSE	1.037 (98.5% of improvement)	Table 6
LOGO CV MSE	2.187 (2.27× standard)	Section 3.8
Design task Spearman ρ	0.363–0.390	Table 9

Model–Representation Interaction. The most striking pattern in the scalar results (Table 1) is the interaction between model type and feature representation: RF is robust but plateaus across all four feature sets, while NN performance varies by 26%. This likely reflects a fundamental difference in how the two models handle high-dimensional inputs: RF's random subspace sampling provides robustness to uninformative dimensions but limits its ability to exploit complex feature interactions. Neural networks can learn nonlinear combinations of PLM embedding dimensions that capture secretion-relevant information, but require sufficient input richness to outperform simpler methods. The poor performance of NN on physicochemical features (MSE 1.328) likely reflects a dimensionality mismatch. With only 156 input dimensions and ∼3000 training samples, the neural network may lack sufficient signal to learn useful representations beyond what the RF already captures through axis-aligned splits. The higher capacity of the NN becomes a liability rather than an asset in this low-dimensional regime.

PLM Model Size. A related question is whether larger PLMs automatically produce better predictions. Surprisingly, the largest PLM (ESM2-3B, 2560 dimensions) does not yield the best results. For RF, ESM2-3B performs worst among all models. For NN, ESM2-3B (MSE 1.067) outperforms ESM2-650M (1.090) but is still outperformed by Ginkgo-AA0 (1.050). With only ∼3000 training samples, the additional dimensions of the 3B model may introduce noise without proportional signal. The Ginkgo-AA0 model, despite having fewer parameters than ESM2-3B, may encode more task-relevant biological information for signal peptide function.

Vector Regression. Beyond the choice of input features, the regression formulation itself also matters. Predicting bin probability distributions instead of scalar WA values yields a 4.7% MSE improvement over the best scalar NN (1.050 → 1.001). This gain likely arises from the richer supervision signal: each training example provides 10 target values instead of one, and the softmax output naturally enforces a valid probability distribution. Focal loss provides a marginal improvement over cross-entropy (+2.3% MSE; Section 3.6), and a controlled comparison across five output formulations confirms that the gain comes primarily from predicting 10-bin distributions rather than from the loss function: the 10-bin CCE model (MSE 0.994) outperforms all classification baselines (MSE 1.13–1.29). However, the vector and scalar NNs also differ in activation function (LeakyReLU vs. ReLU) and the absence of batch normalization (Appendix F), so the improvement cannot be attributed solely to the distributional output without further ablation. With an 80/20 validation split, the best vector model achieves MSE 1.001, above the 0.953 obtained with a near-identical Wolfram Language implementation (same embedding, hidden sizes, dropout, and focal loss; the prior model uses parametric ReLU vs. our LeakyReLU). Removing the validation split and optimizing dropout (0.35 vs. 0.20) with a deeper three-layer architecture (256, 256, 128) yields a 5-seed ensemble MSE of 0.932 (Section 3.4), 2.2% below that benchmark. The key improvements are (i) an additional hidden layer providing greater model capacity, (ii) higher dropout regularization compensating for the absence of early stopping, and (iii) multi-seed ensembling to reduce prediction variance. The 0.953 benchmark was obtained from a single model; our 0.932 result uses a 5-seed ensemble, which inherently reduces prediction variance. Individual seed MSEs for the best configuration range from 0.950 to 1.023 (mean 0.979), so the mean single-seed MSE does not beat 0.953; only one of five seeds individually surpasses it, and the improvement from 0.979 to 0.932 reflects the ensembling benefit.

Cross-Dataset Generalization. While these improvements are substantial within the Grasso dataset, the cross-dataset results reveal a significant limitation. The consistently negative Spearman correlations across all four external datasets are worth examining, though not entirely unexpected. Signal peptide efficiency depends on the cargo protein, promoter, host organism, and growth conditions, all of which differ fundamentally between datasets. The correlations are not merely absent ($\rho \approx 0$) but actively negative, suggesting that the sequence features associated with high secretion in the Grasso system are associated with low performance in other contexts, possibly because the model has learned host- or cargo-specific signal patterns that are maladaptive elsewhere. A signal peptide optimized for β-glucuronidase secretion in *B. subtilis* may be poorly suited for xylanase production in the same organism, let alone for enzyme secretion in *E. coli* or *S. cerevisiae.* The Zhang cross-promoter analysis provides complementary evidence: the same 114 signal peptides correlate at $r = 0.977$ across two promoters, confirming that signal peptide rankings are partially conserved within a system but not across systems. In practice, this means models for signal peptide optimization need to be trained on task-specific data rather than transferred across experimental contexts. Fine-tuning the pretrained vector ensemble on target-domain data provides limited benefit: freezing the learned feature layers and retraining only the output head reverses the negative correlation only for Wu ($-0.206 \rightarrow +0.190$), the sole binary-outcome dataset, but fails to improve or actively worsens performance on all three continuous datasets (Xue: $-0.129 \rightarrow -0.250$; Zhang-P43: $-0.211 \rightarrow -0.271$; Zhang-PglVM: $-0.249 \rightarrow -0.319$). Wu's binary labels may be easier to learn from limited data than continuous activity values, but even the Wu result has high variance ($\pm$ 0.182) and no individual fold reaches significance. These results underscore that transfer learning for signal peptide efficiency remains severely data-limited, and that fine-tuning on small datasets risks overfitting rather than recovering transferable patterns.

Design Task. Despite the limited cross-dataset transferability, the models still prove useful within the Grasso experimental system. The design task demonstrates that models trained on characterized variants can meaningfully rank novel designed signal peptides, with Spearman $\rho = 0.36$–0.39 and classification accuracy of 72–74%, well above the 50% random baseline. Generating ESM2-650M embeddings for the design variants and evaluating all four model–feature combinations reveals that the ESM2-650M NN achieves the highest rank correlation ($\rho = 0.390$), though the improvement over PhysChem features is modest (0.390 vs. 0.378 for NN, 0.363 vs. 0.369 for RF). This contrasts with the larger PLM advantage observed in the scalar regression task, suggesting that the design task's evaluation structure (per-gene ranking rather than absolute WA prediction) may limit the benefit of richer representations. The moderate but significant correlations suggest that these models can serve as a practical filter for prioritizing candidates in experimental screening campaigns.

Limitations. Several methodological caveats apply. The RF and NN hyperparameter searches used different validation strategies (5-fold CV vs. single train/val split), which may introduce some asymmetry in the comparison. While the full-data ensemble MSE

(0.932) falls below the prior benchmark of 0.953, this result relies on a 5-seed ensemble and training without a validation split, which precludes early stopping. The final dropout of 0.35 was selected based on test-set MSE, constituting a form of test-set model selection. Appendix E shows that all dropout values ≥ 0.20 produce statistically indistinguishable validation MSE, so the test-optimal choice is not contradicted by validation data, but neither is it uniquely supported. Validation MSE improves from 1.375 at dropout 0.15 to 1.333–1.347 for all dropout rates ≥ 0.20, with no statistically significant differences among them (all within seed-to-seed variability of $\pm$0.07–0.09); the test-optimal dropout (0.35) produces a validation MSE of 1.347 ± 0.072, indistinguishable from the validation-optimal (0.20, MSE 1.333 ± 0.084). The differences between top-performing models are modest; 95% bootstrap confidence intervals for all models are reported in Appendix D. The cross-dataset significance tests (Table 7) involve eight comparisons; after Bonferroni correction ($\alpha = 0.00625$), all four NN correlations and two of the four RF correlations no longer reach significance. Only the Wu and Xue RF correlations survive correction.

In terms of scope, the dataset is limited to *B. subtilis* with a single reporter enzyme, and Ginkgo-AA0 embeddings—which yielded the best results—were only available for the Grasso dataset; cross-dataset and fine-tuning evaluations use ESM2-650M embeddings, so the actual transferability of Ginkgo-AA0 representations remains untested. The vector regression architecture uses LeakyReLU activation, omits batch normalization, and does not apply L2 kernel regularization, while the scalar NN uses ReLU with batch normalization and L2 regularization; a systematic ablation of these architectural differences was not performed, so the vector regression improvement cannot be attributed solely to the distributional output. The original Grasso train/test partition contains overlapping genes (132 of 134 genes appear in both sets), meaning the model may partly learn gene-specific patterns rather than generalizable signal peptide features. A leave-one-gene-out cross-validation (Section 3.8) addresses this concern.

5 Conclusion

We have systematically evaluated signal peptide efficiency prediction across four feature representations, two model architectures, and two regression formulations, with additional analyses that probe the sources of predictive performance. A vector regression neural network with Ginkgo-AA0 embeddings and focal loss achieves MSE 1.001 ($R^2 = 0.790$) with validation-based training. Full-data optimization with increased dropout and a deeper architecture yields a 5-seed ensemble MSE of 0.932 ($R^2 = 0.804$, Spearman $\rho = 0.873$), improving 23.6% over the physicochemical baseline of 1.22 and falling 2.2% below a prior single-model benchmark of 0.953, although the 95% bootstrap CI [0.823, 1.054] contains the benchmark value and the mean single-seed MSE (0.979) does not surpass it without ensembling.

Two findings temper this headline result. First, a linear probe (single Dense layer) on the same embeddings achieves MSE 1.037, indicating that the PLM embedding encodes the majority of task-relevant information; the MLP architecture contributes a modest further refinement. Second, leave-one-gene-out cross-validation yields MSE 2.19 (Spearman $\rho = 0.720$), a 2.27$\times$ increase over a comparable single-seed standard model (0.964), demonstrating that the overlapping-gene train/test partition substantially inflates apparent performance. The model retains meaningful cross-gene predictive ability ($\rho = 0.720$), but future evaluations should use gene-stratified splits to provide realistic generalization estimates.

Cross-dataset evaluation reveals that zero-shot predictions do not generalize across organisms or experimental systems; fine-tuning the output head reverses the negative correlation for only one of four datasets (Wu, binary outcome), with no individual fold reaching statistical significance. Applied to designed variants, the models achieve classification accuracy of 72–74% and Spearman ρ up to 0.390, demonstrating practical utility for guiding experimental prioritization within the training domain.

Random forests remain competitive and robust across feature types, making them suitable when interpretability or computational efficiency is prioritized. These results demonstrate that PLM embeddings encode rich biological information for signal peptide prediction, but that standard evaluation protocols can substantially overestimate generalization; gene-stratified cross-validation provides a more realistic assessment of model utility. Future work could explore end-to-end fine-tuning of PLM weights on secretion efficiency data, region-aware embedding pooling strategies, or extending the analysis to additional host organisms and cargo proteins.

Data Availability

The original dataset is available as supplementary material to Grasso et al. [2023]. PLM embeddings were generated using publicly available model weights. All analysis code and results are available at `https://github.com/Mehak-W/signal-peptide-experiments`.

References

Stefano Grasso, Valentina Dabene, Margriet M. W. B. Hendriks, Priscilla Zwartjens, René Pellaux, Martin Held, Sven Panke, Jan Maarten van Dijl, Andreas Meyer, and Tjeerd van Rij. Signal peptide efficiency: From high-throughput data to prediction and explanation. *ACS Synthetic Biology*, 12(2):390–404, 2023. doi: 10.1021/acssynbio.2c00328.

Zeming Lin, Halil Akin, Roshan Rao, Brian Hie, Zhongkai Zhu, Wenting Lu, Nikita Smetanin, Robert Verkuil, Ori Kabeli, Yaniv Shmueli, Allan dos Santos Costa, Maryam Fazel-Zarandi, Tom Sercu, Salvatore Candido, and Alexander Rives. Evolutionary-scale prediction of atomic-level protein structure with a language model. *Science*, 379(6637): 1123–1130, 2023. doi: 10.1126/science.ade2574.

Felix Teufel, José Juan Almagro Armenteros, Alexander Rosenberg Johansen, Magnús Halldór Gíslason, Silas Irby Pihl, Konstantinos D. Tsirigos, Ole Winther, Søren Brunak, Gunnar von Heijne, and Henrik Nielsen. SignalP 6.0 predicts all five types of signal peptides using protein language models. *Nature Biotechnology*, 40(7):1023–1025, 2022. doi: 10.1038/s41587-021-01156-3.

Castrense Savojardo, Pier Luigi Martelli, Piero Fariselli, and Rita Casadio. DeepSig: Deep learning improves signal peptide detection in proteins. *Bioinformatics*, 34(10):1690–1696, 2018. doi: 10.1093/bioinformatics/btx818.

Ginkgo Bioworks. AA0: A protein foundation model. Model weights available at `https://models.ginkgobioworks.com`, 2024. Precomputed embeddings used in this study.

Tsung-Yi Lin, Priya Goyal, Ross Girshick, Kaiming He, and Piotr Dollár. Focal loss for dense object detection. In *Proceedings of the IEEE International Conference on Computer Vision*, pages 2980–2988, 2017. doi: 10.1109/ICCV.2017.324.

Zachary Wu, Kevin K. Yang, Michael J. Liszka, Alycia Lee, Alina Batzilla, David Wernick, David P. Weiner, and Frances H. Arnold. Signal peptides generated by attention-based neural networks. *ACS Synthetic Biology*, 9(8):2154–2161, 2020. doi: 10.1021/acssynbio.0c00219.

Songlyu Xue, Xiufang Liu, Yuyang Pan, Chufan Xiao, Yunzi Feng, Lin Zheng, Mouming Zhao, and Mingtao Huang. Comprehensive analysis of signal peptides in Saccharomyces cerevisiae reveals features for efficient secretion. *Advanced Science*, 10(2):2203433, 2023. doi: 10.1002/advs.202203433.

Weiwei Zhang, Mingming Yang, Yuedong Yang, Jian Zhan, Yaoqi Zhou, and Xin Zhao. Optimal secretion of alkali-tolerant xylanase in Bacillus subtilis by signal peptide screening. *Applied Microbiology and Biotechnology*, 100(20):8745–8756, 2016. doi: 10.1007/s00253-016-7615-4.

A Hyperparameter Search Results

Tables 11 and 12 report the top-5 configurations per feature type from the randomized hyperparameter searches (100 RF configs with 5-fold CV; 40 NN configs with 80/20 validation split). Full trial data are available in the GitHub repository.

Table 11: Top-5 random forest configurations per feature type, ranked by 5-fold cross-validation MSE (100 configs searched per type).

Feature	Rank	n_{est}	Depth	Split	Leaf	CV MSE
PhysChem	1	75	None	5	2	2.019
	2	50	25	2	1e-4	2.021
	3	150	30	2	1e-4	2.028
	4	100	30	10	1e-3	2.029
	5	150	25	2	2	2.029
ESM2-650M	1	300	25	1e-3	4	1.843
	2	200	20	2	1e-3	1.845
	3	300	None	10	4	1.845
	4	75	25	5	1e-3	1.847
	5	200	20	2	4	1.847
ESM2-3B	1	150	25	2	2	1.925
	2	300	25	1e-3	4	1.927
	3	100	30	10	1e-3	1.928
	4	200	25	2	1e-3	1.929
	5	200	25	1e-3	1e-3	1.929
Ginkgo-AA0	1	300	None	10	4	1.723
	2	75	None	2	1e-4	1.723
	3	150	30	2	1e-4	1.724
	4	200	20	2	1e-3	1.724
	5	100	25	2	2	1.725

Table 12: Top-5 neural network configurations per feature type (non-log-transformed targets), ranked by validation MSE (40 configs searched per type, 80/20 train/val split). Log-transformed target variants were also evaluated during the search but are excluded here since the optimal skewness was below the transformation threshold (Section 2.2). Full trial data are available in the GitHub repository.

Feature	Rank	Layers	Drop	L2	LR / Batch	Val MSE
PhysChem	1	256, 128	0.3	0.01	1e-3 / 64	1.390
	2	128, 64	0.3	0.01	1e-3 / 32	1.425
	3	128, 64	0.4	1e-4	1e-3 / 64	1.515
	4	128, 64	0.5	0.01	5e-4 / 64	1.529
	5	128, 64	0.4	1e-3	5e-4 / 64	1.551
ESM2-650M	1	256, 128	0.4	0.01	1e-3 / 64	1.241
	2	256	0.3	1e-3	5e-4 / 64	1.266
	3	512, 256	0.5	0.01	5e-4 / 64	1.300
	4	256, 128	0.5	1e-4	5e-4 / 64	1.331
	5	256, 128	0.4	0.01	5e-4 / 32	1.353
ESM2-3B	1	1024, 512	0.4	0.01	1e-3 / 64	1.302
	2	512, 256	0.5	1e-3	1e-3 / 32	1.316
	3	512, 256	0.4	1e-3	5e-4 / 32	1.333
	4	512, 256	0.3	1e-3	1e-3 / 64	1.377
	5	512, 256	0.2	0.01	5e-4 / 32	1.387
Ginkgo-AA0	1	256, 128	0.3	1e-4	5e-4 / 32	1.146
	2	512, 256	0.5	1e-3	5e-4 / 64	1.197
	3	256	0.2	1e-3	1e-3 / 32	1.224
	4	256, 128	0.5	1e-4	1e-3 / 64	1.236
	5	256, 128	0.5	0.01	1e-3 / 32	1.248

B Vector Regression Per-Seed Results

Table 13 reports the per-seed test MSE for all six embedding–loss combinations, along with the ensemble mean.

Table 13: Per-seed test MSE for vector regression models. Each model uses the base architecture (2×256 hidden units, LeakyReLU, 20% dropout, softmax output) with 5 random seeds for train/validation splitting.

Embedding	Loss	Seed 42	Seed 123	Seed 456	Seed 789	Seed 1024	Ensemble
Ginkgo-AA0	Focal	1.026	1.120	1.023	1.128	1.046	**1.001**
Ginkgo-AA0	CE	1.053	1.102	1.054	1.111	1.050	1.003
ESM2-650M	Focal	1.084	1.158	1.100	1.169	1.125	1.049
ESM2-650M	CE	1.084	1.151	1.093	1.194	1.132	1.054
ESM2-3B	CE	1.235	1.224	1.179	1.196	1.214	1.109
ESM2-3B	Focal	1.215	1.220	1.185	1.226	1.193	1.117

C Architecture Search Landscape

Figure 13 shows the full architecture search landscape from the initial full-data vector regression exploration, which identified the (256, 256, 128) three-layer architecture as the top performer prior to the dropout and ensemble optimization in Section 3.4.

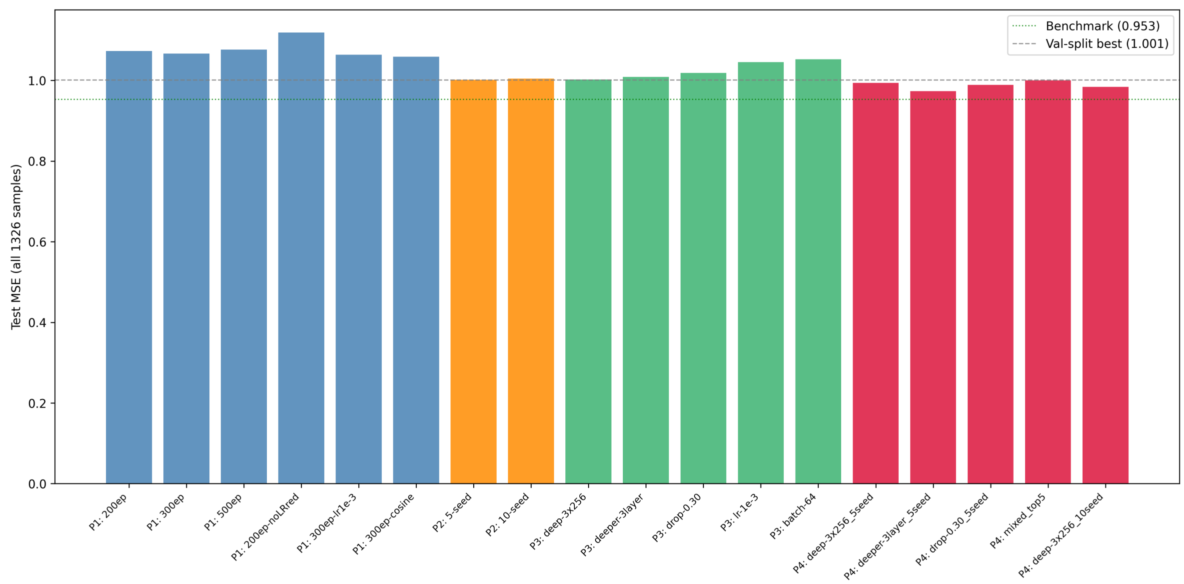

Figure 13: Architecture search results for full-data vector regression (Ginkgo-AA0 + focal loss). Configurations are ranked by test MSE on all 1326 test samples. The three-layer (256, 256, 128) architecture with 5-seed ensemble achieved the lowest MSE (0.973), motivating the subsequent dropout optimization (Section 3.4).

D Bootstrap Confidence Intervals

To quantify uncertainty in test set performance, we compute 95% bootstrap confidence intervals (10,000 resamples) for all 16 models (including the full-data optimized model). Each model is retrained with its best configuration and evaluated on the held-out test set; the test set is then resampled with replacement to obtain the bootstrap distribution. These CIs capture uncertainty due to finite test-set size (sampling variability) but not model uncertainty from random initialization; for ensemble models, the ensemble prediction is fixed before bootstrapping begins.

Table 14: Bootstrap 95% confidence intervals for test MSE, R^2, and Spearman ρ across all models (10,000 bootstrap resamples). †Full-data model: (256, 256, 128) architecture, dropout 0.35, trained on all 3068 samples without validation split, evaluated on all 1326 test samples. *Note:* Point estimates may differ from Tables 1 and 2 because the bootstrap procedure retrains each model with a single random initialization, whereas the main tables report the best configurations selected from hyperparameter searches.

Model	MSE	95% CI	R^2	95% CI	Spearman	95% CI
Random Forest						
RF PhysChem (base)	1.193	[1.078, 1.317]	0.750	[0.724, 0.773]	0.859	[0.844, 0.872]
RF PhysChem (tuned)	1.208	[1.094, 1.328]	0.747	[0.721, 0.770]	0.857	[0.843, 0.870]
RF ESM2-650M	1.189	[1.076, 1.310]	0.750	[0.725, 0.773]	0.853	[0.837, 0.867]
RF ESM2-3B	1.245	[1.126, 1.366]	0.739	[0.713, 0.763]	0.849	[0.833, 0.862]
RF Ginkgo-AA0	1.158	[1.043, 1.281]	0.757	[0.731, 0.781]	0.857	[0.841, 0.870]
Neural Network (scalar)						
NN PhysChem	1.413	[1.253, 1.582]	0.703	[0.666, 0.738]	0.828	[0.809, 0.846]
NN ESM2-650M	1.109	[0.988, 1.240]	0.767	[0.738, 0.794]	0.855	[0.840, 0.869]
NN ESM2-3B	1.175	[1.051, 1.302]	0.753	[0.725, 0.780]	0.850	[0.835, 0.864]
NN Ginkgo-AA0	1.191	[1.076, 1.317]	0.750	[0.723, 0.774]	0.848	[0.833, 0.861]
Vector regression (5-seed ensemble)						
Vec ESM2-650M CE	1.051	[0.937, 1.175]	0.780	[0.753, 0.804]	0.861	[0.845, 0.875]
Vec ESM2-650M focal	1.045	[0.934, 1.167]	0.781	[0.755, 0.804]	0.860	[0.844, 0.874]
Vec ESM2-3B CE	1.122	[1.007, 1.247]	0.765	[0.737, 0.789]	0.857	[0.841, 0.870]
Vec ESM2-3B focal	1.091	[0.981, 1.211]	0.771	[0.745, 0.794]	0.859	[0.844, 0.872]
Vec Ginkgo-AA0 CE	0.995	[0.887, 1.112]	0.791	[0.766, 0.814]	0.867	[0.852, 0.880]
Vec Ginkgo-AA0 focal	0.997	[0.887, 1.115]	0.791	[0.765, 0.814]	0.866	[0.850, 0.880]
Full-data optimized (5-seed ensemble, no val split)						
Vec Ginkgo-AA0 focal†	**0.932**	[0.823, 1.054]	**0.804**	[0.779, 0.828]	**0.873**	[0.857, 0.887]

Figure 14 presents these intervals as a forest plot, showing the relative performance and uncertainty of all 16 models.

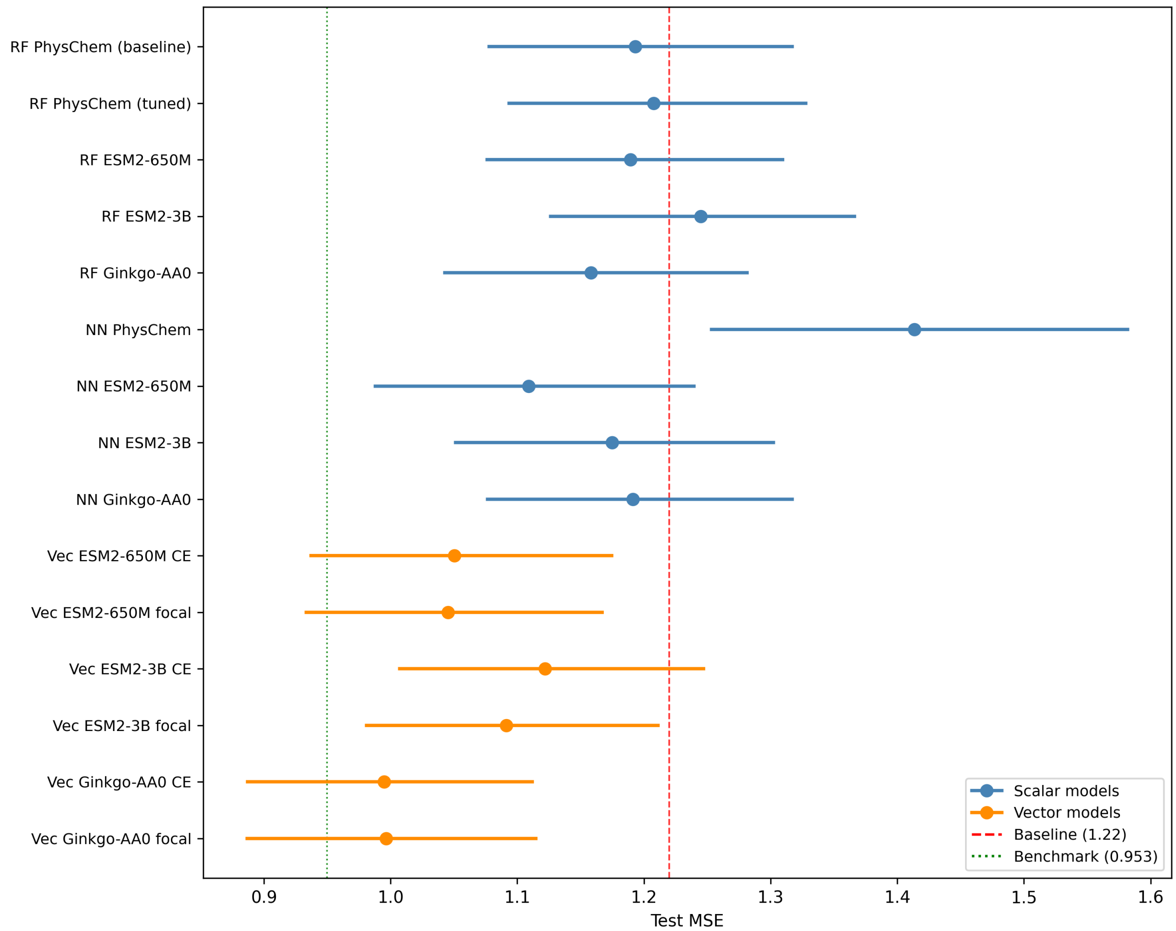

Figure 14: Forest plot of test MSE with 95% bootstrap confidence intervals for all 16 models. Blue markers indicate scalar models (RF and NN); orange markers indicate vector regression ensembles. Dashed red line: physicochemical baseline (MSE 1.22); dotted green line: Dr. Schrier benchmark (MSE 0.95).

E Dropout Validation

To confirm that the dropout selection in Section 3.4 is not an artifact of test-set model selection, we repeated the dropout sweep using an independent 80/20 train/validation split from the training data. For each dropout rate in $\{0.15, 0.20, 0.25, 0.30, 0.35\}$, we trained 5-seed ensembles on 80% of the training data and evaluated on the held-out 20% validation set. The same (256, 256, 128) architecture, focal loss, and training protocol from Section 3.4 were used.

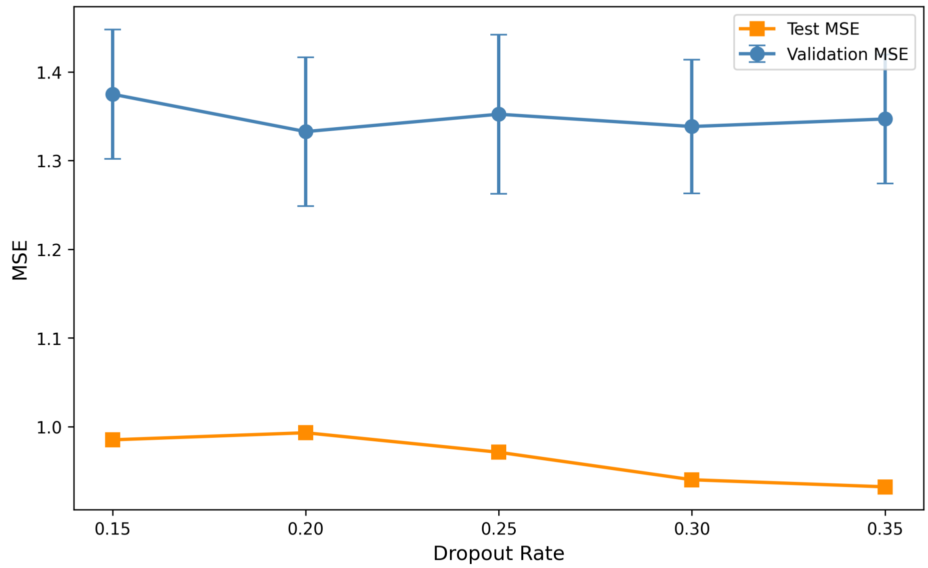

Figure 15: Dropout validation: MSE on held-out validation data (80/20 split, blue) versus test data (full-data training from Section 3.4, orange). The validation curve is essentially flat for dropout ≥ 0.20, while the test curve shows clearer improvement at higher dropout. Error bars on validation MSE represent standard deviation across 5 seeds. Absolute validation MSE is higher because models train on only 80% of the data.

As shown in Figure 15, validation MSE drops from 1.375 at dropout 0.15 to a minimum of 1.333 at dropout 0.20, then remains between 1.338 and 1.352 for dropout 0.25–0.35; all values at dropout ≥ 0.20 are within seed-to-seed variability ($\pm$0.07–0.09). The test-optimal dropout (0.35, validation MSE 1.347 ± 0.072) and the validation-optimal (0.20, 1.333 ± 0.084) are statistically indistinguishable, as are all intermediate values. This supports the conclusion that dropout values in the range 0.20–0.35 produce equivalent generalization performance, and the test-optimal choice of 0.35 is not contradicted by the validation data. The higher absolute MSE on validation ($\sim$1.35 vs. $\sim$0.93 on test) is expected because validation models train on only 80% of the data (2454 vs. 3068 samples).

F ReLU-Squared Activation Comparison

We compared ReLU-squared ($\text{ReLU}^2(x) = \max(0, x)^2$) against the default LeakyReLU activation in the best vector regression architecture (256, 256, 128; dropout 0.35; focal loss; 5-seed ensemble). LeakyReLU achieves MSE 0.961 [0.847, 1.085], while ReLU-squared achieves MSE 5.11 [4.78, 5.44], a catastrophic degradation ($R^2 = -0.073$) worse than a mean-prediction baseline (MSE 4.76). The large per-seed variance (4.67–9.07) indicates training instability rather than a systematic accuracy–sparsity tradeoff. ReLU-squared does produce substantially sparser internal representations (Hoyer sparsity 0.67–0.92 and 56–75% exact zeros vs. 0.27–0.32 and 0% zeros for LeakyReLU), but this comes at the cost of model collapse rather than a graceful tradeoff. This experiment uses only two learning rates (5×10^{-4} with fallback to 2×10^{-4}) and does not include BatchNorm, gradient clipping, or standard

ReLU controls; the catastrophic failure may therefore reflect a hyperparameter mismatch rather than a fundamental limitation of ReLU-squared. Figure 16 shows the comparison.

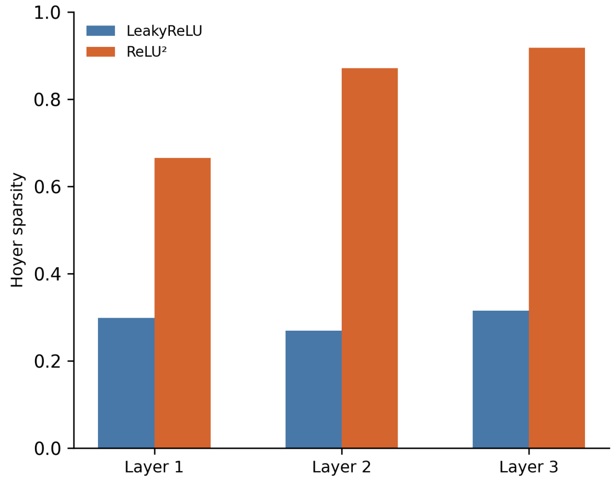

Figure 16: Hoyer sparsity per hidden layer for ReLU-squared vs LeakyReLU. ReLU-squared produces 2–3× sparser representations but degrades prediction accuracy from MSE 0.961 to 5.11.